

Exercise Sheet 6

Efficient Computing/Parallel Computing 2 in SS 2026

Date of issue: 19.5.2026

Deadline: 26.5.2026

The exercises marked with ★ are relevant for admission to the examination and must be submitted in electronic form in Moodle by the submission deadline.

Exercise 6.1 (*OpenMP-parallelized matrix-vector operations*).

Develop parallel versions of the following C/C++ code fragments by augmenting them with OpenMP directives. Try to reduce the number of implicit barriers and specify every variable as **shared** or **private**. Implement the parallel versions in C/C++. Write a test bench to validate the correctness of the implementation and measure the run-times. Compare the run-times of both versions, the serial version and the parallel version, for each of the given code fragments using suitable problem sizes and a varying number of threads. Report and explain the results.

File: Code-fragment1.c ➡

```
a)   for (i=0; i<m; i++){
      y[i]=0;
      for (j=0; j<n; j++){
        y[i]= y[i] + A[i][j]*x[j];
      }
    }
```

File: Code-fragment2.c ➡

```
b)   s=0;
      for (i=0; i<m; i++){
        y[i]=0;
        for (j=0; j<n; j++){
          y[i]= y[i] + A[i][j]*x[j];
        }
        s += y[i];
      }
```

File: Code-fragment3.c ➡

```
c)   y[0]=0;
      for (i=1; i<m; i++){
        y[i]=y[i-1];
        for (j=0; j<n; j++){
          y[i]= y[i] + A[i][j]*x[j];
        }
      }
```